Propositional Logic

Consider the following argument:

1. *Either this man is dead or my watch has stopped.*
2. *My watch is still ticking.*
3. *Therefore, this man is dead.*

Here, we ***deduce***, or ***infer***, the truth of a ***conclusion*** – a particular ***proposition***, or ***statement*** (that a man is dead) – under the assumption that two other propositions (***premises***) are true.

This argument is logically valid (even if it is based on a nonsense premise).

Failures in such arguments underpin the majority of software system failures.

The role of ***Propositional Logic*** is to formalise such reasoning to ensure only valid arguments can be made.

Propositions / Statements

Propositions, or ***statements***, are declarations which are either *true* or *false*.

The following are all propositions:

- $2 + 3 = 5$.
- $2 + 3 = 6$.
- Joel didn't do his homework.
- What Felix says is false.

The following are **not** propositions:

- $2 + x = 6$. *(predicated on the value of x)*
- Do your homework, Joel! *(command)*
- Is there life on Mars? *(question)*
- What this sentence says is false. *(paradox)*

Deductions / Inferences

A ***deduction***, or ***inference***, is when you deduce, or infer, one statement (a ***conclusion***) from one or more other statements (***premises***).

Valid deductions

1

All men have green blood.
Socrates is a man.
Therefore, Socrates has green blood.

2

Some animals are mammals.
All mammals lay eggs.
Therefore, some animals lay eggs.

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Some even numbers are > 5
Some numbers > 5 are odd
Therefore some even numbers are odd
```

Invalid deductions

3

All men have green blood.
Socrates has green blood.
Therefore, Socrates is a man.

4

Some animals are mammals.
Some mammals lay eggs.
Therefore, some animals lay eggs.

The Language of Propositional Logic

Statements (formulas) of Propositional Logic are similar to algebraic formulas, like $x^2 + 2y - 15$.

- Algebraic formulas evaluate to *numerical values* based on the values of the numerical variables in the formula.
- Propositional formulas evaluate to *truth values* based on the values of the propositional variables in the formula.

For ***propositional variables*** we will always use upper-case letters or (more commonly) meaningful words starting with an upper-case letter.

Example: Let Dead represent the statement: *"This man is dead"*.

Algebraic formulas are built up using four basic operations:

addition ($+$); *subtraction* ($-$); *multiplication* ($\times$); *division* ($\div$).

Propositional formulas use five basic ***propositional connectives (operations)***:

not ($\neg$); *and* ($\wedge$); *or* ($\vee$); *implies* ($\Rightarrow$); *is equivalent to* ($\Leftrightarrow$).

Negation

connective: negation

symbol: $\neg \mathrm{p}$

pronounced: *"not* p*"*

In English:

- *not* p *(or, rather, statement* p *with "not" after verb)*
- p *does not hold / is not true / is false*
- *it is not the case that* p

Example: If Dead stands for *"This man is dead"*, then $\neg \mathrm{Dead}$ says

- *This man is not dead*
- *It is not the case that this man is dead*

Negation

Exercise: Rewrite the following statements without negations at the start.

1. $\neg$ *"The Earth revolves around the sun"*
2. $\neg$ *"All of my children are boys"*
3. $\neg\,(2+2 \leq 4)$ 2 + 2 > 4

$2+2 \not\leq 4$



Observation:

If a proposition is not true, then it must be false.

Conversely, if it is not false, then it must be true.

In particular, if a proposition is *not* not true, then it is true:

$\neg\neg\,\mathrm{p}$ is equivalent to p

This is known as the *Law of Double Negation*

Conjunction

connective: conjunction

symbol: $\mathrm{p} \wedge \mathrm{q}$ (p and q are called *conjuncts*)

pronounced: "p and q"

In English:

- p *and* q
- p *but* q
- *not only* p *but also* q

Example: If Dead stands for *"This man is dead,"* and $\mathrm{Stopped}$ stands for *"My watch has stopped"*, then $\mathrm{Dead} \wedge \mathrm{Stopped}$ says

- *This man is dead* and *my watch has stopped*
- *Not only is this man dead, but so is my watch*

Disjunction

connective: disjunction

symbol: $p \vee q$ (p and q are called *disjuncts*)

pronounced: "p or q"

In English:

- p *or* q
- p *or* q *or both*
- p *and/or* q
- p *unless* q

Example: If Dead stands for *"This man is dead"* and $\mathrm{Stopped}$ stands for *"My watch has stopped"*, then $\mathrm{Dead} \vee \mathrm{Stopped}$ says

- *Either this man is dead or my watch has stopped*
- *If this man is alive, then my watch must have stopped*

(Possibly the man is dead *and* the watch has stopped.)

Conjunction and Disjunction

Exercise: Are the following disjunctions true or false?

C1. $(3 < 2) \wedge (3 < 5)$ (F, F, T)

C2. $(5 < 4) \wedge (7 < 5)$ (F, F, F)

C3. $(5 < 6) \wedge (6 < 8)$ (T, T, T)

D1. $(3 < 2) \vee (3 < 5)$ (F, T, T)

D2. $(5 < 4) \vee (7 < 5)$ (F, F, F)

D3. $(5 < 6) \vee (6 < 8)$ (T, T, T)



Observation:

As $\neg p$ is true whenever p is *not* true,

$p \wedge \neg p$ will always be false, and

$p \vee \neg p$ will always be true.

(The latter is known as the *Law of Excluded Middle*.)

Inclusive vs Exclusive Or

The disjunction operation $\vee$ is always used in an ***inclusive*** sense, in that it is true if ***one or both*** disjuncts are true. For example:

> *Joel came in last place in the round-robin competition;*
> *thus, either Felix beat him or Oskar beat him.*
>
> (It may be the case that both Felix and Oskar beat Joel.)

Sometimes an ***exclusive*** interpretation is intended when we use the word "or", whereby we allow ***only one*** of the disjuncts to be true. For example:

> *The light is either on or off.*
>
> *You can either have coffee or tea with your meal.*

This kind of disjunction, written $\oplus$, is called ***exclusive or***.

Implication

connective: implication

symbol: $p \Rightarrow q$

pronounced: "p implies q"

In English:

- p *implies* q
- *if* p *then* q
- q *if* p
- p *only if* q
- q *whenever* p
- p *is a sufficient condition for* q
- q *is a necessary condition for* p

Example: If $\mathrm{SignalRed}$ stands for *"The signal shows red"* and $\mathrm{TrainStop}$ stands for *"The train stops"*, then $\mathrm{SignalRed} \Rightarrow \mathrm{TrainStop}$ says

If the signal shows red then the train stops

The only scenario in which this statement is false is if the signal shows red but the train does not stop.

Hence the statement does not contradict the situation in which the signal does not show red and yet the train nevertheless stops.

Implication



Exercise: Let JoelHappy stand for *"Joel is happy"* and let AmandaHappy stand for *"Amanda is happy"*

Each of the three statements below translates as either

$\text{JoelHappy} \Rightarrow \text{AmandaHappy}$ or as $\text{AmandaHappy} \Rightarrow \text{JoelHappy}$

Determine which in each case.

1. *Joel is happy whenever Amanda is happy* `AmandaHappy ⇒ JoelHappy`
2. *Joel is happy only if Amanda is happy* `JoelHappy ⇒ AmandaHappy`
3. *Joel is happy unless Amanda is not happy*

`"X unless Y" ~` $\neg Y \Rightarrow X$ `so:` $\neg\neg$`AmandaHappy ⇒ JoelHappy` `so:` `AmandaHappy ⇒ JoelHappy`

Exercise: On the door of a house is the following warning to potential thieves:

`Bark` `Bite`

Barking dogs don't bite ... but my dog doesn't bark !

`Bark ⇒ ¬Bite` `¬Bark`

`No, not necessarily!`

Should a potential thief *necessarily* be concerned?

`AmandaHappy ⇒ JoelHappy`

`"X unless Y" ~` $\neg X \Rightarrow Y$ `so:` `¬JoelHappy ⇒ ¬AmandaHappy` `so:`

P, Q Statements

$$P \Rightarrow Q \qquad\qquad \neg Q \Rightarrow \neg P$$

if P then Q
Q if P
P only if Q

1. Joel is happy only if Amanda is Happy

   Joel is only happy if Amanda is Happy

2. I will only trust you if you did not lie to me about the cake

   I will trust you only if you did not lie to me about the cake

3. You will only pass the exam if you understand propositional logic

   You will pass the exam only if you understand propositional logic

Equivalence

connective: equivalence

symbol: $\mathrm{p} \Leftrightarrow \mathrm{q}$

pronounced: "p if, and only if, q"

In English:

- p *if, and only if,* q
- p *is equivalent to* q
- p *is a necessary and sufficient condition for* q

Example: Let

- $\mathrm{TrainEnter}$ stand for *"The train enters the tunnel"* and
- $\mathrm{TunnelClear}$ stand for *"The tunnel is clear"*

Then $\mathrm{TrainEnter} \Leftrightarrow \mathrm{TunnelClear}$ says

The train enters the tunnel if, and only if, the tunnel is clear

This statement is false if one of these two scenarios holds:

- the train enters the tunnel while the tunnel is not clear;
- the tunnel is clear but the train does not enter.

Equivalent Propositions

Propositions can be re-written – *carefully!* – without changing their meaning.

Example: $p \Leftrightarrow q$ is equivalent to $(p \Rightarrow q) \wedge (q \Rightarrow p)$

Saying

p *is true if, and only if,* q *is true*

is the same as saying

if p *is true then* q *is true and if* q *is true then* p *is true*

Example: $p \Rightarrow q$ is equivalent to $\neg p \vee q$

Saying

if p *is true then* q *is true*

is the same as saying

either p *is false or* q *is true*